

GenAI Engineer - Take Home Assessment

Overview

This take-home assessment will evaluate your technical proficiency, problem-solving skills, and ability to work with generative AI technologies. You will be tasked with developing a basic chatbot that demonstrates agentic behavior and incorporates Retrieval-Augmented Generation (RAG) functionality. The focus is on well-written, tested, and maintainable code. Please complete the assessment and submit your solution within one week.

Guidelines

1. **Submission Format:** Provide your code, documentation, and run instructions in a single zip file or via a GitHub (private) repository.
2. **Programming Languages:** Python is required.
3. **Documentation:** Ensure your code is well-documented and includes instructions on how to run it end-to-end. Include a concise report in PDF or Markdown format detailing your approach, challenges faced, and results.
4. **Evaluation Criteria:**
 - a. Required functionality is implemented as specified in the document.
 - b. Code quality and adherence to software engineering best practices.
 - c. The quality of documentation and report.

Assessment Task - Agentic Movie Recommendation Chatbot

You are tasked with developing a prototype for a Movie Recommendation Chatbot that demonstrates agentic behavior and incorporates Retrieval-Augmented Generation (RAG) functionality.

The chatbot should be capable of responding to two types of queries about movie recommendations using external data. Each one of the queries should be handled by **two specialized agents**.

Agents LLM Models

We provide an API Gateway to interact with OpenAI models. The API key will be provided to you attached to this assessment. To set it up, use the usual environment variable in your setup:

```
OPENAI_API_KEY=<your-api-key>
```

Agent 1: Current Trending Movies

Data Source

Use the TMDB ([themoviedb.org](https://developer.themoviedb.org/reference/trending-movies)) data to get the current trending movies through this API endpoint: <https://developer.themoviedb.org/reference/trending-movies>.

You can create a profile and get an API Key for free with an excellent rate-limiting policy (50 requests per second). Check their [FAQ](#) page for more info. TMDB exposes a lot of data and more endpoints. To reduce the scope of this assessment, the agent **must only use the trending-movies endpoint**.

Queries Examples

1. "Which recent movie do you recommend to watch today?"
2. "What is the current best movie released lately?"
3. "Is there a good movie to watch about superheroes lately?"

Agent 2: Netflix Movies and Shows

Data Source

Use this Kaggle dataset of Netflix TV shows and movies with descriptions: <https://www.kaggle.com/datasets/victorsoeiro/netflix-tv-shows-and-movies>.

Queries Examples

1. "Find me a movie to watch on the TV about action and spies"
2. "What are some good Netflix nature documentaries?"

3. “I want to see a romantic comedy movie. What do you recommend?”

Agent Routing

- Implement a mechanism to **route queries to the appropriate agent based on the content of the query**.
- Ensure each agent can handle and respond to queries within its domain of expertise.

User Interface

- There are no requirements on the user interface or API. It is up to you to decide the implementation based on your available time and experience. Some perfectly valid examples of interfaces for users are:
 - Command line scripts or command line interactive interface.
 - REST API.
 - Simple web-based UI.
- Independently of the user interface, the user experience must be as good as possible under all circumstances. So the interaction with the chatbot has the following requirements:
 - The chatbot **must respond in streaming** so the users can see answers as soon as possible.
 - The user interface should be as simple as possible and require minimal setup from a user perspective.
 - Implement mechanisms that hide errors and failures from the users.

Software Engineering Practices

- Write clean, maintainable, modular, and well-documented code.
- Follow best practices for software development.
- Make sure the code is packaged appropriately so it can be run in a fresh environment.
- Include automated tests or step-by-step instructions to test the chatbot end-to-end.
- Unit tests are not required but are highly encouraged.

Documentation and Report

- Include detailed instructions on how to run the code end to end (including tests and evaluation).
- Write a brief report detailing:
 - Your approach, challenges faced, and results.
 - Explanation of the design choices and methodology, including selection of tools and models.
 - Next steps, productization strategy, and improvements

Additional Notes

- Feel free to use any open-source software and tools.
- Feel free to make reasonable assumptions but document them clearly. Interpretation of the assessment is part of the assessment evaluation process.